```java
import java.io.BufferedReader;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import java.util.Iterator;
import java.util.LinkedHashMap;

public class MacroP1 {

    public static void main(String[] args) throws IOException{
        BufferedReader br=new BufferedReader(new FileReader("macro_input.asm"));

        FileWriter mnt=new FileWriter("mnt.txt");
        FileWriter mdt=new FileWriter("mdt.txt");
        FileWriter kpdt=new FileWriter("kpdt.txt");
        FileWriter pnt=new FileWriter("pntab.txt");
        FileWriter ir=new FileWriter("intermediate.txt");
        LinkedHashMap<String, Integer> pntab=new LinkedHashMap<>();
        String line;
        String Macroname = null;
        int mdtp=1,kpdtp=0,paramNo=1,pp=0,kp=0,flag=0;
        while((line=br.readLine())!=null)
        {

            String parts[]=line.split("\\s+");
            if(parts[0].equalsIgnoreCase("MACRO"))
            {
                flag=1;
                line=br.readLine();
                parts=line.split("\\s+");
                Macroname=parts[0];
```

```java
if(parts.length<=1)
{
   mnt.write(parts[0]+"\t"+pp+"\t"+kp+"\t"+mdtp+"\t"+(kp==0?kpdtp:(kpdtp+1))+"\n");
   continue;
}
for(int i=1;i<parts.length;i++) //processing of parameters
{
   parts[i]=parts[i].replaceAll("[&,]", "");
   //System.out.println(parts[i]);
   if(parts[i].contains("="))
   {
      ++kp;
      String keywordParam[]=parts[i].split("=");
      pntab.put(keywordParam[0], paramNo++);
      if(keywordParam.length==2)
      {
         kpdt.write(keywordParam[0]+"\t"+keywordParam[1]+"\n");
      }
      else
      {
         kpdt.write(keywordParam[0]+"\t-\n");
      }
   }
   else
   {
      pntab.put(parts[i], paramNo++);
      pp++;
   }
}
mnt.write(parts[0]+"\t"+pp+"\t"+kp+"\t"+mdtp+"\t"+(kp==0?kpdtp:(kpdtp+1))+"\n");
kpdtp=kpdtp+kp;
```

```java
    //System.out.println("KP="+kp);
  }
  else if(parts[0].equalsIgnoreCase("MEND"))
  {
    mdt.write(line+"\n");
    flag=kp=pp=0;
    mdtp++;
    paramNo=1;
    pnt.write(Macroname+":\t");
    Iterator<String> itr=pntab.keySet().iterator();
    while(itr.hasNext())
    {
      pnt.write(itr.next()+"\t");
    }
    pnt.write("\n");
    pntab.clear();
  }
  else if(flag==1)
  {
    for(int i=0;i<parts.length;i++)
    {
      if(parts[i].contains("&"))
      {
        parts[i]=parts[i].replaceAll("[&,]", "");
        mdt.write("(P,"+pntab.get(parts[i])+")\t");
      }
      else
      {
        mdt.write(parts[i]+"\t");
      }
    }
```

```java
                mdt.write("\n");
                mdtp++;
            }
            else
            {
                ir.write(line+"\n");
            }
        }
        br.close();
        mdt.close();
        mnt.close();
        ir.close();
        pnt.close();
        kpdt.close();
        System.out.println("MAcro PAss1 Processing done. :)");
    }

}

import java.io.BufferedReader;
import java.io.FileReader;
import java.io.FileWriter;
import java.util.HashMap;
import java.util.Vector;

public class MacroP2 {

    public static void main(String[] args) throws Exception {
        BufferedReader irb=new BufferedReader(new FileReader("intermediate.txt"));
        BufferedReader mdtb=new BufferedReader(new FileReader("mdt.txt"));
        BufferedReader kpdtb=new BufferedReader(new FileReader("kpdt.txt"));
```

```java
        BufferedReader mntb=new BufferedReader(new FileReader("mnt.txt"));

        FileWriter fr=new FileWriter("pass2.txt");

        HashMap<String, MNTEntry> mnt=new HashMap<>();
        HashMap<Integer, String> aptab=new HashMap<>();
        HashMap<String,Integer> aptabInverse=new HashMap<>();

        Vector<String>mdt=new Vector<String>();
        Vector<String>kpdt=new Vector<String>();

        int pp,kp,mdtp,kpdtp,paramNo;
        String line;
        while((line=mdtb.readLine())!=null)
        {
            mdt.addElement(line);
        }
        while((line=kpdtb.readLine())!=null)
        {
            kpdt.addElement(line);
        }
        while((line=mntb.readLine())!=null)
        {
            String parts[]=line.split("\\s+");
            mnt.put(parts[0], new MNTEntry(parts[0], Integer.parseInt(parts[1]),
    Integer.parseInt(parts[2]), Integer.parseInt(parts[3]), Integer.parseInt(parts[4])));

        }

        while((line=irb.readLine())!=null)
        {
```

```
String []parts=line.split("\\s+");
if(mnt.containsKey(parts[0]))
{
    pp=mnt.get(parts[0]).getPp();
    kp=mnt.get(parts[0]).getKp();
    kpdtp=mnt.get(parts[0]).getKpdtp();
    mdtp=mnt.get(parts[0]).getMdtp();
    paramNo=1;
    for(int i=0;i<pp;i++)
    {
        parts[paramNo]=parts[paramNo].replace(",", "");
        aptab.put(paramNo, parts[paramNo]);
        aptabInverse.put(parts[paramNo], paramNo);
        paramNo++;
    }
    int j=kpdtp-1;
    for(int i=0;i<kp;i++)
    {
        String temp[]=kpdt.get(j).split("\t");
        aptab.put(paramNo,temp[1]);
        aptabInverse.put(temp[0],paramNo);
        j++;
        paramNo++;
    }

    for(int i=pp+1;i<parts.length;i++)
    {
        parts[i]=parts[i].replace(",", "");
        String splits[]=parts[i].split("=");
        String name=splits[0].replaceAll("&", "");
        aptab.put(aptabInverse.get(name),splits[1]);
```

```
    }
    int i=mdtp-1;
    while(!mdt.get(i).equalsIgnoreCase("MEND"))
    {
        String splits[]=mdt.get(i).split("\\s+");
        fr.write("+");
        for(int k=0;k<splits.length;k++)
        {
            if(splits[k].contains("(P,"))
            {
                splits[k]=splits[k].replaceAll("[^0-9]", "");//not containing number
                String value=aptab.get(Integer.parseInt(splits[k]));
                fr.write(value+"\t");
            }
            else
            {
                fr.write(splits[k]+"\t");
            }
        }
        fr.write("\n");
        i++;
    }

    aptab.clear();
    aptabInverse.clear();
}
else
{
    fr.write(line+"\n");
}
```

```java
    }

    fr.close();
    mntb.close();
    mdtb.close();
    kpdtb.close();
    irb.close();
    }
}

public class MNTEntry {
    String name;
    int pp, kp, mdtp, kpdtp;
    public MNTEntry(String name, int pp, int kp, int mdtp, int kpdtp) {
        super();
        this.name = name;
        this.pp = pp;
        this.kp = kp;
        this.mdtp = mdtp;
        this.kpdtp = kpdtp;
    }
    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public int getPp() {
        return pp;
    }
    public void setPp(int pp) {
```

```java
        this.pp = pp;
    }
    public int getKp() {
        return kp;
    }
    public void setKp(int kp) {
        this.kp = kp;
    }
    public int getMdtp() {
        return mdtp;
    }
    public void setMdtp(int mdtp) {
        this.mdtp = mdtp;
    }
    public int getKpdtp() {
        return kpdtp;
    }
    public void setKpdtp(int kpdtp) {
        this.kpdtp = kpdtp;
    }
    // Added main method for testing
    public static void main(String[] args) {
        MNTEntry entry = new MNTEntry("ExampleMacro", 1, 2, 3, 4);
        System.out.println("Name: " + entry.getName());
        System.out.println("PP: " + entry.getPp());
        System.out.println("KP: " + entry.getKp());
        System.out.println("MDTP: " + entry.getMdtp());
        System.out.println("KPDT: " + entry.getKpdtp());
    }
}
```